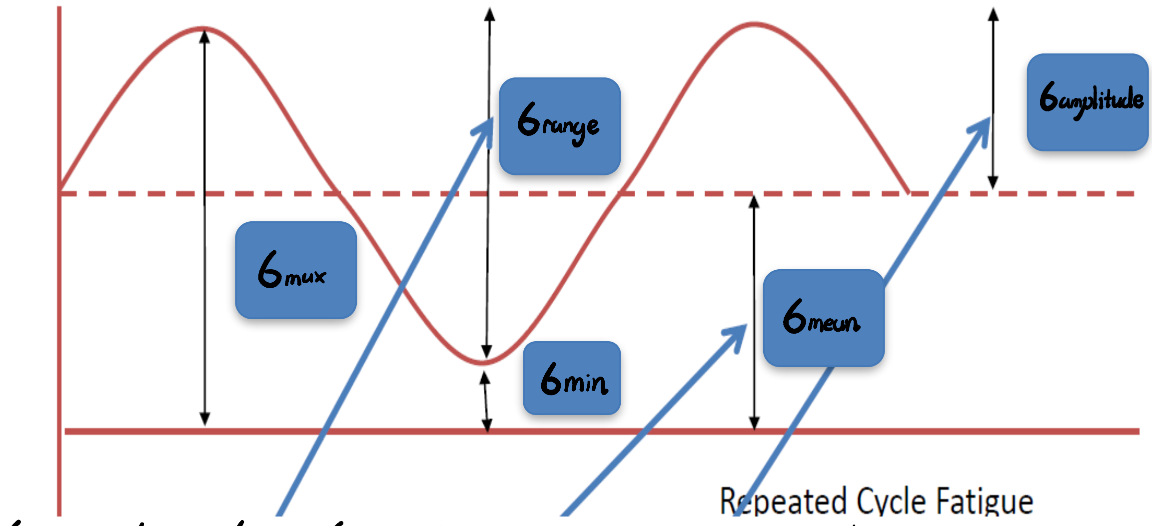
3. Failure: Fatigue

1. Definitions of terms in fatigue

Mechanical components can fail at stresses well below the tensile stress of the material if subjected to cyclic loading (alternating stress and strain). This type of material failure is fatigue.

Mark σ_{max}, σ_{min}, σ_{mean}, $\sigma_{amplitude}$, σ_{range} in the figure below. And write the equation to calculation σ_{mean}, $\sigma_{amplitude}$, σ_{range} by knowing σ_{max} and σ_{min}.



$\sigma_{range} = \sigma_{max} - \sigma_{min}$ $\sigma_{mean} = \frac{1}{2}(\sigma_{max} + \sigma_{mean})$ $\sigma_{amplitude} = \frac{1}{2}(\sigma_{max} - \sigma_{min})$

Stress ratio R $=\sigma_{min}/\sigma_{max}$, R= -1 when alternating load around zero mean stress.

There are two different types of fatigue, Low Cycle Fatigue (LCF) and High Cycle Fatigue (HCF). HCF is for fatigue life > 10^4 cycles and it is under relatively low $\sigma_{amplitude}$ (stress lower than yield stress); LCF is for fatigue life <10^4 cycles and it is under relatively high $\sigma_{amplitude}$ (stress higher than yield stress).

2. S-N curve

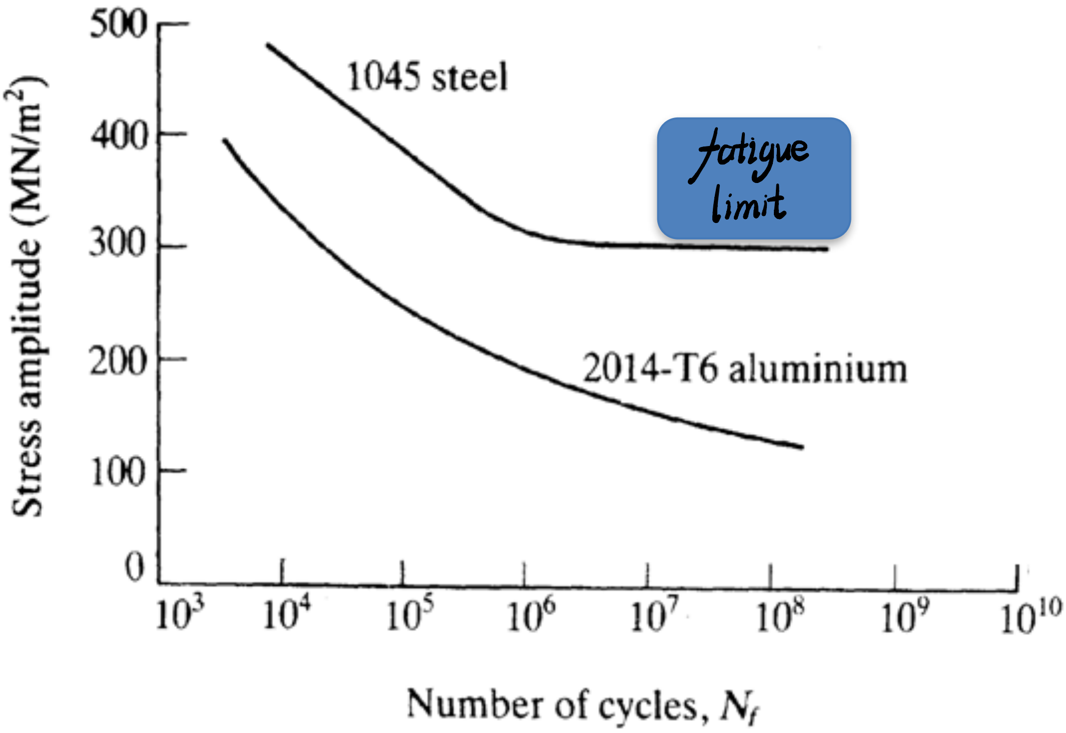
Fatigue data are typically shown as S-N curve. Some materials such as steel have a fatigue limit (mark it in the curve below as well), below which the material has an "infinite" lifetime. For materials such as Al without fatigue limit, often endurance limit σ_e is defined.

For example, at 10^7 cycles, σ_e = 150 MPa.

Lifetime decreases with increases stress amplitude.

S-N curve usually is for 0 mean stress. But real world components often have non zero mean stress. The higher the mean stress, the shorter fatigue life.

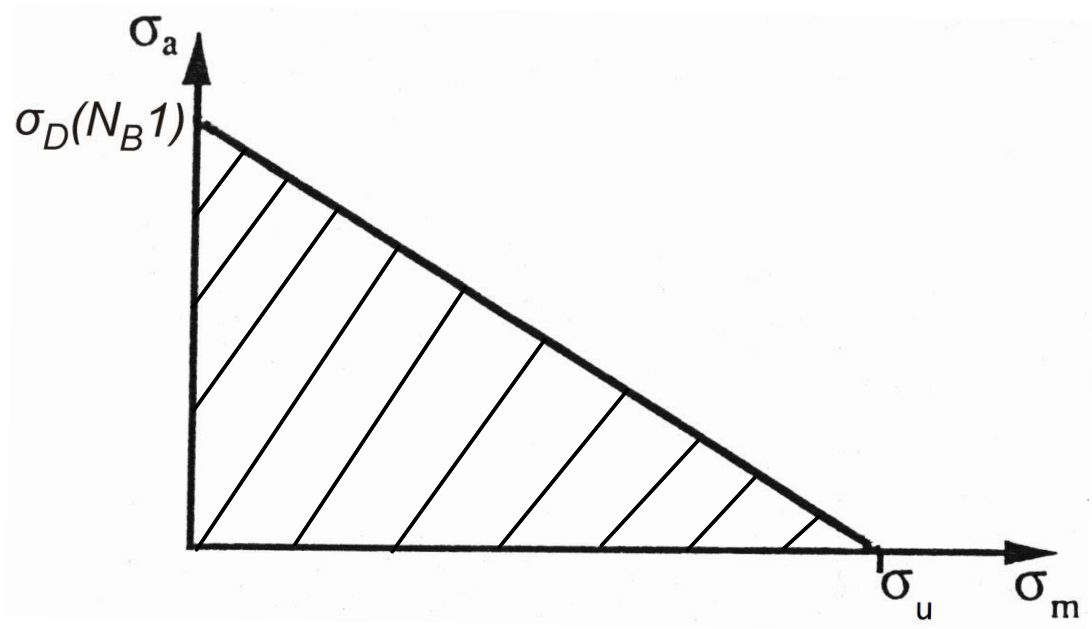
- Tensile stresses lead to shorter life times (crack opening)
- Compression to an longer in fatigue life (crack closing)



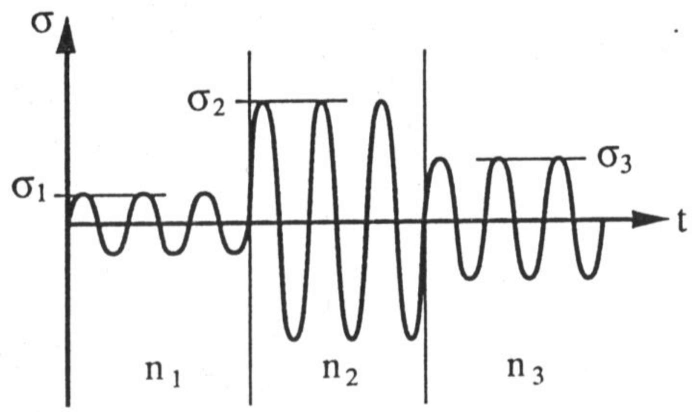
Read the S-N curve of 2014-T6 alloys:
N_B (100MPa) = 5×10^{9};
N_B (180 MPa) = 10^6;
N_B (300MPa) = 2×10^{4};

3. **Goodman law**

Goodman law shows empirical relation to extrapolate fatigue life from testing data at $\sigma_{mean} = 0$. Goodman relation can be used to draw lines of constant lifetime. Please mark the safe region in the Goodman law below (see Tutorial 3 for calculation using Goodman law):



4. **Palmgren-Miner-Rule**
   Palmgren-Miner-Rule assumes linear damage accumulation, giving possibility to interpret complex real world data using controlled lab generated fatigue data.

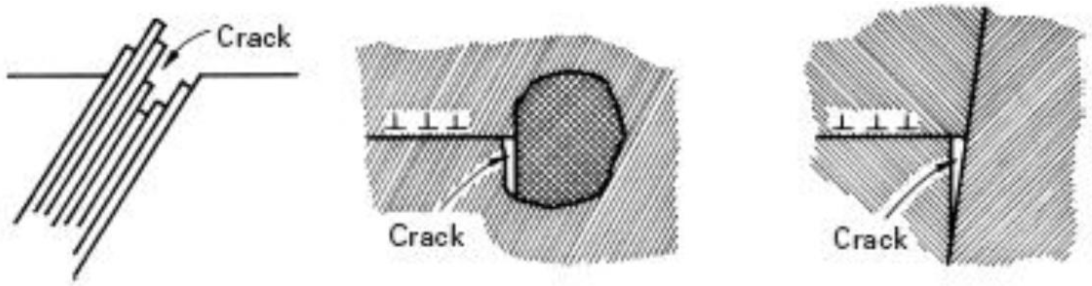


Use Palmgren-Miner-Rule to indicate in which condition, the component under cyclic stresses in the figure above won't have fatigue failure. $N_1(\sigma_1)$ can be used to present the number of fatigue life cycles N_1 under applied stresses amplitude σ_1.

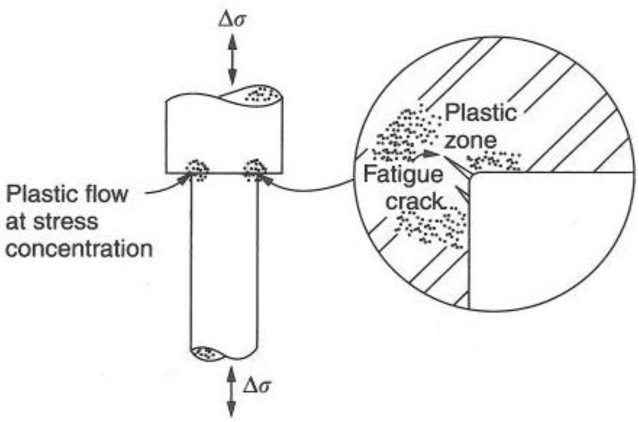
$$\frac{n_1}{N_1(\sigma_1)} + \frac{n_2}{N_2(\sigma_2)} + \frac{n_3}{N_3(\sigma_3)} < 1$$

5. **Fatigue crack initiation.**

For LCF, the sources of cracks can be slip band, inclusion and grain boundary. For slip band, plastic deformation roughens the surface within a small number of cycles. Slip planes cause sub-micrometre extrusion and intrusion at the surface.



For HCF, there is no bulk plastic deformation so almost all of the life is taken up in initiating the crack. There is some local plasticity in regions where a notch or scratch or change in section concentrate the stress. Cracks eventually form in these regions.



6. **Fatigue growth**

The cylic stress intensity factor $\Delta K = Y\Delta\sigma\sqrt{\pi a}$. As the crack grows, ΔK increase. Thus, ΔK increases with time.

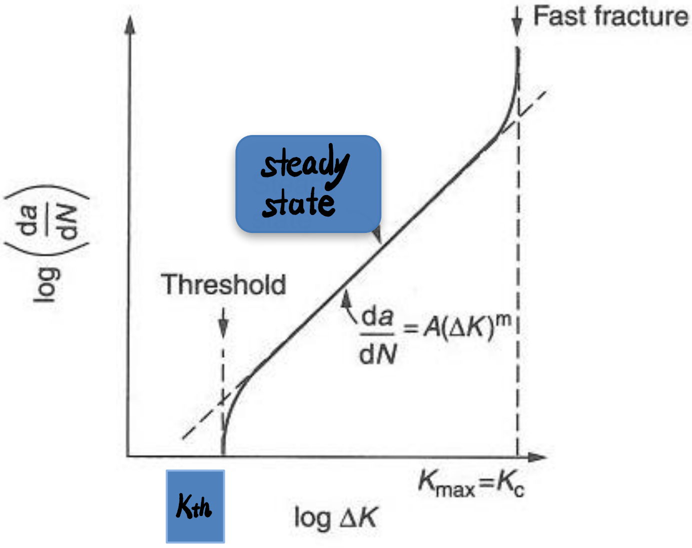
Fatigue crack growth rate da/dN.

Mark ΔK_{th} in the figure below. ΔK_{th} is ΔK threshold value, below which, crack is not advancing.

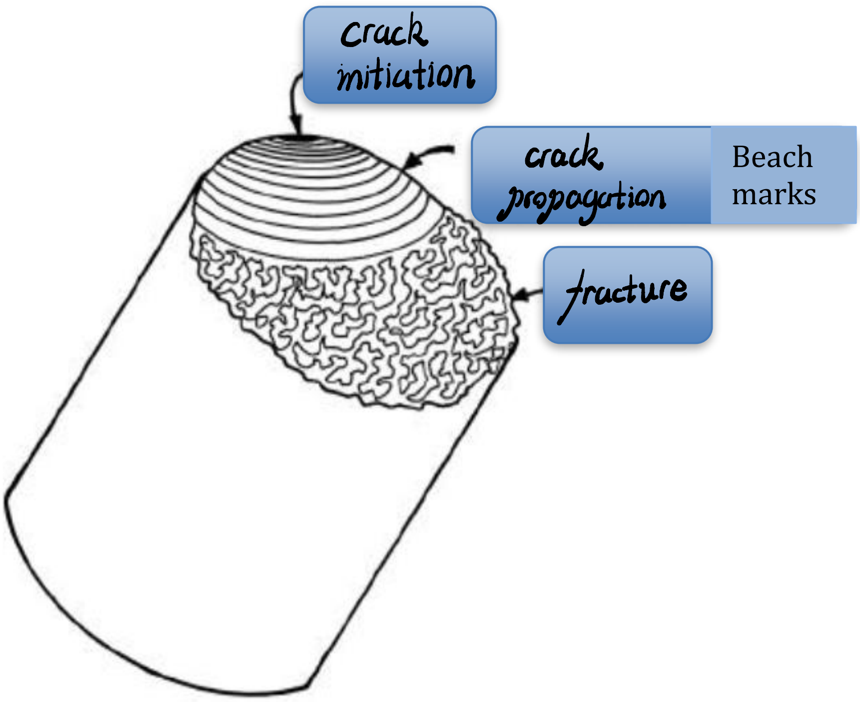
Mark the steady-state regime in the figure below. In this region the Paris Law is applicable, which can be used to predict the fatigue life by knowing a_0 (initial crack size), a_f (critical crack size) and the stress change $\Delta\sigma$ is valid. See Tutorial 3 for calculation examples.

$$N_f = \frac{1}{AY^m \Delta\sigma^m \pi^{\frac{m}{2}}} \int_{a_0}^{a_f} \frac{da}{a^{\frac{m}{2}}}$$

$$a_f = \frac{1}{\pi}\left(\frac{K_c}{Y\sigma_{max}}\right)^2$$



7. **Fatigue fracture surface**.
Label the typical regions.



In which the beach marks indicate that the material has failed by fatigue and each striation is produced by a single stress cycle.

8**. Design against fatigue**

For fail safe design, we can compare σ_e (endurance limit) to ΔK_{th}, design below lower one. However, in this case the components are normally too big or heavy and it is not realistic especially for lightweight application.

For limited lifetime design which is usually used in plane design, Paris law is commonly used for pre-existing crack and for determining necessary inspection intervals.